

VIGNESHWARAN C

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Summary

CS undergraduate building **AI agents and secure backends**: tool-calling agent loops over local LLMs, auth/RBAC for multi-tenant SaaS, and CI security pipelines. Merged open-source contributor and active CTF player.

Education

St. Joseph's College of Engineering

B.E. in Computer Science and Engineering – CGPA: 8.13

Chennai, India

2024 – 2028

IIT Madras

B.S. in Data Science and Applications (Dual Degree)

Online / Chennai, India

2024 – Present

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, C++, Java, Dart, Bash

AI & Agents: ReAct Loops, Tool Calling, Ollama, LangChain, FAISS, RAG, MediaPipe, OpenCV, Scikit-learn

Backend & Web: FastAPI, Node.js, Express, PostgreSQL, Redis, Firebase, React, Redux Toolkit, Flutter, Tailwind CSS

Security & DevOps: GitHub Actions, Docker, Terraform (Azure), Semgrep, Gitleaks, Trivy, OWASP ZAP, nmap, BlackArch

Professional Experience

Application Development Intern

ATRIBS Global Technology Solutions

Jun – Jul 2025 | Dec 2025 – Jan 2026

Chennai • second term offered on merit

- Built **Police Akka** (citizen SOS app + police dashboard) with Firebase real-time streams, live GPS tracking and a Groq-hosted LLM for legal guidance.
- Presented the architecture to Rockwell Automation mentors and folded their review into the data-flow design.

AI Automation & Data Science Intern

Toeddro Adventures Pvt. Ltd.

Apr – Jul 2026

Remote

- Built a resumable 3-stage Python lead pipeline (URL discovery, site scraping, domain-level dedup), collecting ~450 unique operator leads.
- Built **Swift Mailer**, a Node.js outreach tool with CSV import, SMTP checks and rate-limit-safe send delays.

Projects

Jarvis-Linux – Voice-Driven Desktop Agent | *Python, Ollama, FAISS, psutil, Hyprland*

- **ReAct loop** over a local LLM (Llama 3.2 via Ollama) choosing from 26 system tools (apps, files, audio, psutil telemetry, Hyprland workspaces); tool errors are fed back so the model retries.
- Two-tier memory: a 10-turn window plus long-term semantic recall with **FAISS** over MiniLM embeddings, exposed as remember/recall/forget tools.

Law-Suite – Multi-Tenant Legal SaaS (team; I own auth, RBAC, audit) | *React, Node.js, Express, PostgreSQL*

- Auth for a mentor-led 14-module platform: bcrypt, 6-digit OTP (5-min expiry), JWT refresh-token rotation and login rate limiting (5 / 10 min).
- 4-tier RBAC (SuperAdmin to Client) enforced per endpoint in Express middleware, with an immutable audit trail.

NeuroMotion AI – Motion Analysis (UIP 17 shortlist, team of 4; now solo research) | *FastAPI, MediaPipe, DTW, Flutter*

- Tracks 33 pose landmarks, reduces them to 6 joint angles, and aligns free throws to an expert reference with **DTW**; a 10-rule engine gives corrections.
- Built with a Unisys mentor (out after round 2); continuing solo toward a research paper and patent.

Security & Open Source

- **Open source (OSCG 2026):** 2 merged PRs: bounded GitHub API use in hiero-bot-py (pagination caps, 5-way concurrency, TTL cache) and lock-duration bounds in Truxify's AtomicSwap contract; 3 more in review.
- **DevSecOps pipeline** (task-manager): CI gates on Semgrep, Gitleaks, dependency audits and Trivy; OIDC + Terraform deploy to Azure, then OWASP ZAP.
- **CTF:** Flag Hunt 2026, team CYBORK (full clear; I solved the web set incl. SQLi auth bypass) | Bandit 0–20 writeups
- **Problem solving:** LeetCode 239 solved (contest rating 1,614, top 23%) | SkillRack 2,169 solved

Certifications

- **Google Cloud:** AI Agents with ADK, MLOps for GenAI
- **Microsoft Applied Skills:** GitHub Copilot, GenAI in Azure
- **PostgreSQL**
- **Oracle:** OCI 2025 Foundations
- **NPTEL:** Python for Data Science (Elite + Silver), Cloud Computing
- **Cisco:** Intro to Cybersecurity
- **MongoDB,**
- HackerRank** skill badges

Achievements & Leadership

- **Winner**, REC Titanium 2026 (Smart City Debug War Room) | **Finalist**, StartUp TN Tourism Hackathon
- 3rd Prize, Spark Sync (SJCE) | 4th Place, Codeathon (CIT) | Technical Lead, college symposium (200+ attendees)